

8. Scaling an image to its Bigger and Smaller sizes. AIM:-Converting an image to image to its Bigger and Smaller sizes.

```
# Install OpenCV
```

```
!pip install opencv-python
```

```
# Import libraries
```

```
import cv2
```

```
import matplotlib.pyplot as plt
```

```
from google.colab import files
```

```
# Upload image
```

```
uploaded = files.upload()
```

```
# Get uploaded image name
```

```
image_name = list(uploaded.keys())[0]
```

```
# Read image
```

```
image = cv2.imread(image_name)
```

```
# Check if image is loaded
```

```
if image is None:
```

```
    print("Error: Image not loaded!")
```

```
else:
```

```
    # Convert BGR to RGB for displaying
```

```
    original = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
```

```
# Get original dimensions
```

```
height, width = image.shape[:2]
```

```
# Bigger Image (3x)
```

```
bigger_image = cv2.resize(image, (int(width*3), int(height*3)))
```

```
# Smaller Image (0.5x)
```

```
smaller_image = cv2.resize(image, (int(width*0.5), int(height*0.5)))
```

```
# Convert to RGB
```

```
bigger_rgb = cv2.cvtColor(bigger_image, cv2.COLOR_BGR2RGB)
```

```
smaller_rgb = cv2.cvtColor(smaller_image, cv2.COLOR_BGR2RGB)
```

```
# Save images
```

```
cv2.imwrite("Bigger_Image.jpg", bigger_image)
```

```
cv2.imwrite("Smaller_Image.jpg", smaller_image)
```

```
# Display Original Image
```

```
plt.figure(figsize=(5,5))
```

```
plt.imshow(original)
```

```
plt.title("Original Image")
```

```
plt.axis("off")
```

```
plt.show()
```

```
# Display Bigger Image
```

```
plt.figure(figsize=(7,7))
```

```
plt.imshow(bigger_rgb)
```

```
plt.title("Bigger Image")
```

```
plt.axis("off")
```

```
plt.show()
```

```
# Display Smaller Image
```

```
plt.figure(figsize=(3,3))  
plt.imshow(smaller_rgb)  
plt.title("Smaller Image")  
plt.axis("off")  
plt.show()
```

```
# Download output images
```

```
files.download("Bigger_Image.jpg")
```

```
files.download("Smaller_Image.jpg")
```

Original Image



Bigger Image



Smaller Image



9. Perform Rotation of an image to clockwise and counter clockwise direction. AIM:- Rotation of an image to clockwise and counter clockwise direction.

```
# Install OpenCV
```

```
!pip install opencv-python
```

```
# Import libraries
```

```
import cv2
```

```
import matplotlib.pyplot as plt
```

```
from google.colab import files
```

```
# Upload image
```

```
uploaded = files.upload()
```

```
# Get uploaded image name automatically
```

```
image_name = list(uploaded.keys())[0]
```

```
# Read image
```

```
img = cv2.imread(image_name)
```

```
# Check if image is loaded
```

```
if img is None:
```

```
    print("Error: Image not loaded!")
```

```
else:
```

```
    # Rotate clockwise (90°)
```

```
clockwise = cv2.rotate(img, cv2.ROTATE_90_CLOCKWISE)

# Rotate counter-clockwise (90°)
counter_clockwise = cv2.rotate(img, cv2.ROTATE_90_COUNTERCLOCKWISE)

# Save output images
cv2.imwrite("Clockwise_Rotation.jpg", clockwise)
cv2.imwrite("CounterClockwise_Rotation.jpg", counter_clockwise)

# Convert BGR to RGB for display
original = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
clockwise_rgb = cv2.cvtColor(clockwise, cv2.COLOR_BGR2RGB)
counter_rgb = cv2.cvtColor(counter_clockwise, cv2.COLOR_BGR2RGB)

# Display Original Image
plt.figure(figsize=(6,6))
plt.imshow(original)
plt.title("Original Image")
plt.axis("off")
plt.show()

# Display Clockwise Rotated Image
plt.figure(figsize=(6,6))
plt.imshow(clockwise_rgb)
plt.title("Clockwise Rotation (90°)")
plt.axis("off")
plt.show()

# Display Counter-Clockwise Rotated Image
```

```
plt.figure(figsize=(6,6))  
plt.imshow(counter_rgb)  
plt.title("Counter-Clockwise Rotation (90°)")  
plt.axis("off")  
plt.show()  
  
# Download output images  
files.download("Clockwise_Rotation.jpg")  
files.download("CounterClockwise_Rotation.jpg")
```

Original Image



Clockwise Rotation (90°)



Counter-Clockwise Rotation (90°)



10. Perform moving of an image from one place to another. AIM:- Moving of an image from one place to another.

Install OpenCV

```
!pip install opencv-python
```

Import libraries

```
import cv2
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
from google.colab import files
```

Upload image

```
uploaded = files.upload()
```

```
# Get uploaded image name
image_name = list(uploaded.keys())[0]

# Read image
image = cv2.imread(image_name)

# Check if image is loaded
if image is None:
    print("Error: Image not loaded!")
else:
    # Convert BGR to RGB
    original = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

    # Get image dimensions
    rows, cols = image.shape[:2]

    # Translation matrix (Move 100 pixels right and 50 pixels down)
    M = np.float32([[1, 0, 100],
                    [0, 1, 50]])

    # Move (translate) the image
    moved_image = cv2.warpAffine(image, M, (cols, rows))

    # Convert to RGB for display
    moved_rgb = cv2.cvtColor(moved_image, cv2.COLOR_BGR2RGB)

    # Save output image
    cv2.imwrite("Moved_Image.jpg", moved_image)
```

```
# Display Original Image
```

```
plt.figure(figsize=(6,6))
```

```
plt.imshow(original)
```

```
plt.title("Original Image")
```

```
plt.axis("off")
```

```
plt.show()
```

```
# Display Moved Image
```

```
plt.figure(figsize=(6,6))
```

```
plt.imshow(moved_rgb)
```

```
plt.title("Moved Image")
```

```
plt.axis("off")
```

```
plt.show()
```

```
# Download output image
```

```
files.download("Moved_Image.jpg")
```

Original Image



Moved Image

